Company

MTH 501 QUIZ #3

GUjjar gujjar

Remember me in your prayers. It's my humble request

Please If any mistake forgive me

MTH501 - Linear Algebra (Quiz 3)

Question # 9 of 10 (Start time: 10:20:32 PM, 30 January 2023)

Two vectors u and v in R^n are orthogonal if ------

Select the correct option

- $u.v \neq 0$
- $u.v = 0$



Question # 10 of 10 (Start time: 10:27:37 PM, 30 January 2023) Total Marks: 1

Let u, v and w be vectors in R^n, then

Select the correct option

Reload Math Equations

- $(u+v).w=u.w+v.w-2u.w$
- None of the above
- $(u+v).w=u.w+v.w$
- $(u+v).w=u.w-v.w$

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 10:30:53 PM, 30 January 2023)

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix i

Select the correct option

○	Blocked rectangular
○	Blocked diagonal-constant
○	Blocked square
○	Blocked identity

Click t

Question # 8 of 10 (Start time: 10:31:56 PM, 30 January 2023)

Suppose that real solutions y_1 and y_2 of $x' = Ax$, form a basis for the two-

Select the correct option

○	Linearly Independent
◉	Linearly dependent

sec(s)

MTH501 - Linear Algebra (Quiz 3) Quiz Start Time: 10:32 PM, 30 January 2023

Question # 2 of 10 (Start time: 10:33:52 PM, 30 January 2023) Total Marks: 1

An n x n matrix A is _____ if and only if A has n linearly independent vectors.

Select the correct option

- symmetric
- scalar
- diagonalizable
- singular

Click to Save Answer & Move to Next Question

BC200402750: MUHAMMAD TAHIR

MTH501 - Linear Algebra (Quiz 3)

Question # 5 of 10 (Start time: 01:04:41 PM, 30 January 2023)

A vector whose length is 1 is called ----------

Select the correct option

- (●) Unit Vector
- ( ) Identity Vector

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MTH501 - Linear Algebra (Quiz 3) Quiz Start Ti

Question # 9 of 10 (Start time: 01:09:20 PM, 30 January 2023)

The _____ *vector is orthogonal to every vector in* R^n

Select the correct option

- unit
- normalized
- none of these
- zero



Click to Save Ans

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MTH501 - Linear Algebra (Quiz 3)

Question # 10 of 10 (Start time: 01:10:23 PM, 30 January 2023)

Two vectors u and v are orthogonal to each other if _____.

Select the correct option

●	u + v = 0
○	u . v = 0
○	u . v = 1
○	u - v = 0

MTH501 - Linear Algebra (Quiz 3) Quiz start time: 01:46 PM, 30 January 2023

Question # 7 of 10 (Start time: 01:48:40 PM, 30 January 2023) **Total Marks:** 1

The process of creating the unit vector u from v sometime called---------

Select the correct option

Reload Math Equations

- Normalizing v
- Diagonalizing v

Click to Save Answer & Move to Next Question

Time Left

MTH501 - Linear Algebra (Quiz 3)

Quiz Start Time: 01:30 PM, 30 January 2

Question # 8 of 10 (Start time: 01:38:11 PM, 30 January 2023)

Total Mar

When a matrix A has a complex eigenvalue with positive real part, the trajectories spiral ----------

Select the correct option

Reload Math Equation

Inwards

Outwards



Click to Save Answer & Move to Next Question

uestion # 7 of 10 (Start time: 01:36:39 PM, 30 January 2023)

For any vectors u and v, the length of vector u − v will be _____.

elect the correct option

- ○ ‖ u − v ‖
- ○ | u − v |
- ○ ‖ u . v ‖
- ○ u . v

Click to Save Answer & M

Definition

Two vectors in u and v in R^n are orthogonal (to each other) if $u.v = 0$

Note

The zero vector is orthogonal to every vector in R^n because $0^t.v = 0$ for all v in R^n.

The Pythagorean Theorem

Example 5

If $u=(u_1,u_2,u_3)$ and $v=(v_1,v_2,v_3)$, then

$$dist(u,v)=\|u-v\|=\sqrt{(u-v).(u-v)}$$

$$=\sqrt{(u_1-v_1)^2+(u_2-v_2)^2+(u_3-v_3)^2}$$

Definition

Two vectors in u and v in R^n are orthogonal (to each other) if $u.v=0$

Note

The zero vector is orthogonal to every vector in R^n because $0^t.v=0$ for all v in R^n.

The Pythagorean Theorem

Two vectors u and v are orthogonal if and only if $\|u+v\|^2=\|u\|^2+\|v\|^2$

Orthogonal Complements

Question # 1 of 10 (Start time: 02:14:33 PM, 30 January 2023)

Two vectors u and v are orthogonal to each other if _____.

Select the correct option

○	$u + v = 0$
○	$u \cdot v = 1$
○	$u - v = 0$
○	$u \cdot v = 0$



Click

Question # 5 of 10 (Start time: 02:19:38 PM, 30 January 2023)

A matrix $[A]_{n \times n}$ has both positive and negative eigenvalues so in this case origin behaves as a ----------

Select the correct option

Reload

- Saddle point
- Critical point

Click to Save Answer & Move to

If a vector x_0 is specified, then the initial value problem is to construct the unique function x such that --------

Select the correct option

Re

- $x' = Ax$ and $x(0) = x_0$
- $x' = Ax'$ and $x(0) = x_0$

Click to Save Answer & Mo

uestion # 3 of 10 (Start time: 02:17:55 PM, 30 January 2023)

Let V be a one Eigen vector then conjugate eigen vector is represented by,

elect the correct option

- $\widetilde{V}$
- $\overline{V}$
- V
- V'

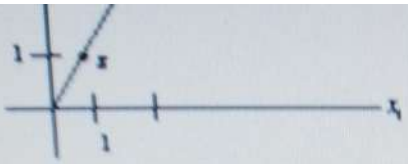
Click to Sav

Figure 2 Normalizing a vector to produce a unit vector.

Definition

For u and v vectors in R^n, the distance between u and v, written as dist (u, v), is the length of the vector $u - v$. That is

$$dist(u, v) = \|u - v\|$$



(b)

Figure 2 Normalizing a vector to produce a unit vector.

Definition

For u and v vectors in R^n, the distance between u and v, written as dist (u, v), is the length of the vector $u - v$. That is

$$dist(u, v) = \|u - v\|$$

Example 4

Compute the distance between the vectors $u = (7, 1)$ and $v = (3, 2)$

Solution

$$u - v = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 7-3 \\ 1-2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\text{dist}(u, v) = \|u - v\| = \sqrt{(4)^2 + (-1)^2} = \sqrt{16+1} = \sqrt{17}$$